Now, p is about as concentrated as it can be, but for an ordinary macroscopic object, the uncertainty relation is pretty much saturated—the left-hand side is roughly *equal* to $\hbar$. The reasons for this are very complicated, and I won't go into them here. Instead, let's assume this is true and work out the implications. What is Δp? It's $m\Delta v$, which gives us

$$m\Delta v\Delta x \sim \hbar.$$

Rearranging the symbols, we can then write

$$\Delta v\Delta x \sim \frac{\hbar}{m}$$

or

$$\Delta x \sim \frac{\hbar}{m\Delta v}.$$

Now, if I put a bowling ball on the ground, I know very well that the uncertainty in its velocity is not very big. As the ball gets heavier and heavier, you might expect the uncertainty in velocity to get smaller and smaller. But, in any case, the right-hand side has an m in the denominator, and regardless of Δv, as m gets smaller, Δx will get bigger. And in particular, it will tend to get bigger than the features in the potential.

In the quantum mechanical limit where m is very small and Δx tends to be big, the wave function will move under the influence of a ragged potential, which it sees as being much sharper and more featured than the wave function itself. That's when the wave function breaks up. On the other hand, as m gets very large, Δx gets small. For a large bowling ball, the wave packet might be *very* concentrated. When it moves through a spiky potential, this *tiny* wave function encounters a potential whose features are (comparatively) very broad. Moving through broad smooth features does not break the wave function into pieces. Large masses and smooth potentials characterize the